



FEL2 – Soil/Water Model Tank



The apparatus consists of a narrow transparent fronted tank, which may be partially filled with any soil type. It is supported on a bench mounted frame. The equipment is self-contained with a sump tank, pump and flow meter fitted behind the tank.

PRODUCT CODE: FEL2

Categories: Educational, FE Series - Irrigation Water Management, FEL2 - Soil/Water Model Tank

Tags: Drip Points , Drip Rate , FE Series , FEL2 , Flood , Infiltration , Irrigation , Irrigation Water Management , Laboratory , Land Slopes , Small Scale , Soil Profile , Soil Surface , Soil/Water Model Tank , Students , Sub-Surface , Surface Run-Off , Tile Drain , Trickle Irrigation , Water Movement , Wetted Profile

DESCRIPTION

The Surface Irrigation Apparatus, FEL2 has been developed to help students of irrigation understand more fully the interaction of factors which influence water movement both on the soil surface and in the soil profile.

The equipment allows actual surface irrigation experiments to be performed on a small scale in the laboratory.

The apparatus consists of a narrow transparent fronted tank, which may be partially filled with any soil type. It is supported on a bench mounted frame. The equipment is self-contained with a sump tank, pump and flow meter fitted behind the tank.

Water may be discharged along the soil surface or directed to twin drip nozzles to demonstrate trickle irrigation. Measurement of rates of water penetration into the soil is aided by a grid on the tank front.

Removable end plates enable soil samples to be changed quickly and easily.

TECHNICAL SPECIFICATIONS

FEATURES & BENEFITS

Demonstration Capabilities

- Understand surface and sub-surface effects of surface water application

- Understand optimum irrigation application rates to maximise infiltration and minimise surface run-off
- Demonstrating the effect of rate of discharge on the advance and infiltration of water into the soil
- Demonstrating the effect of soil texture on the advance and infiltration of water into the soil
- Demonstrating the effect of land slopes on the advance and infiltration of water into the soil
- Demonstrating the effect of surface irregularities on the advance and infiltration of water into the soil
- Demonstrating the use of drip irrigation methods and how drip rate and spacing of drip points affects the wetted profile within the soil
- Demonstrating by visualising the flow lines how a tile drain works

Ordering specification

Soil/Water Model Tank comprising:

- Sand tank, 1.00m x 0.025m x 0.450m (L x W x H)
- One large side formed from a sheet of transparent material
- Metal frame supporting sump tank from which water is pumped via a flow meter to the soil surface
- Flood and drip surface outlets (two can be used together by fitting a Y-connector)
- Overflow system to remove surface water to sump
- Complete with instruction manual

Requirements

Electrical supply:

FEL2-A: 220-240V/1Ph/50Hz

FEL2-B: 120V/1Ph/60Hz

FEL2-G: 220-240V/1Ph/60Hz

Cold water supply

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

Volume: 0.9m³

Gross Weight: 130kg

Dimensions

Length: 1.1m

Width: 0.5m

Height: 0.75m

Order Codes

- **FEL2-A:** 220-240V/1Ph/50Hz
- **FEL2-B:** 120V/1Ph/60Hz
- **FEL2-G:** 220-240V/1Ph/60Hz



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